

EC101

Post Midsem Part

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Chapter 1

Foreword?

I have tried to convert most of what Sir said in lectures (Post-midsem) into notes. There is still room for error. Apologies in advance for any mistakes (hopefully few to none) made. This is my interpretation of what we are taught. Also, sir is very energetic, and as a consequence, there may be a lack of a 'coherent' flow of thought in these notes. This should hopefully have every lecture? Thanks to the ppl who helped me out in this :D I don't really know how useful this is but here goes :)



Figure 1.1: Stonks.

Chapter 2

Syllabi?

Please do check the moodle (additional references have been provided by sir)

Quiz II Syllabus

Rudiger Dornbusch, Stanley Fischer and Richard Startz; 11th Edition.

Part 1 : Chapter 1 and Chapter 2

Part 4 : Chapter 13

Part 5 : Chapter 19

N. Gregory Mankiw, *Principles of Macroeconomics*; 3rd Edition.

Chapter 10

Chapter 11

Chapter 14

Figure 2.1: Syllabus for Quiz 2

Chapter 3

Lecture 1

Microeconomics is the study of decision making at the margin. Macroeconomics, in contrast, looks at the aggregate behavior of the whole economy. Instead of one consumer or one firm, it studies totals like national income, total output (GDP), inflation, unemployment, interest rates, and economic growth.

Consider the very simplistic model where the population grows at the rate of 'n per person'. This way, if the population is denoted by L , we have:

$$\frac{dL}{dt} = nL$$

This differential equation has the solution:

$$L = L_0 e^{nt}$$

As can be seen from the exponential term, this population grows at an explosive rate (assuming n is positive here). A larger population competing for nearly the same amount of resources causes a growth in scarcity, which percolates into the society over time.

The direction the economy takes is governed by three questions:

- What to produce?
- How to produce it?
- For whom is this being produced?

If Capital is denoted by K , and Labour is denoted by L , then an industry with a higher ratio of K to L is called more **capital intensive**. i.e. for two industries, $A(K_1, L_1)$ and $B(K_2, L_2)$, if

$$\frac{K_1}{L_1} > \frac{K_2}{L_2}$$

then we say that industry A is more capital intensive than industry B .

Macroeconomics can be thought of as a study of the structure and performance of national economies, and of the policies that the government and/or policy makers implement to influence the overall economic performance.

Macroeconomic Theory is the theory of National Income, Employment, Prices and Money. It has orientations towards both theory and policies. The following are a few terms and variables that Macroeconomic Theory deals with:

- Total output in economy
- Aggregate price level
- Employment (or Unemployment)
- Interest Rate
- Wage rate
- Exchange rate
- Labour Market
- Debt
- International trades, (exports and imports)

The trap of hyperinflation can make countries poor.

Functions of a Government

- Defence
- Judiciary
- Enterprises
- Social Welfare
- Infrastructure
- Control of Monopoly
- Social Justice
- Macroeconomic Policy

Consider the following diagram (This is probably an Edgeworth Box.)

In this diagram, O_A is a poor individual while O_B is a rich individual. Notice that even if shifting of the endowment point towards a more 'equal' distribution of resources seems good for social welfare, it isn't possible to do so, because O_B has earned this positioning of the endowment point. Even if this box were to expand, it would expand in the favour of O_B , since he/she/they too pay taxes

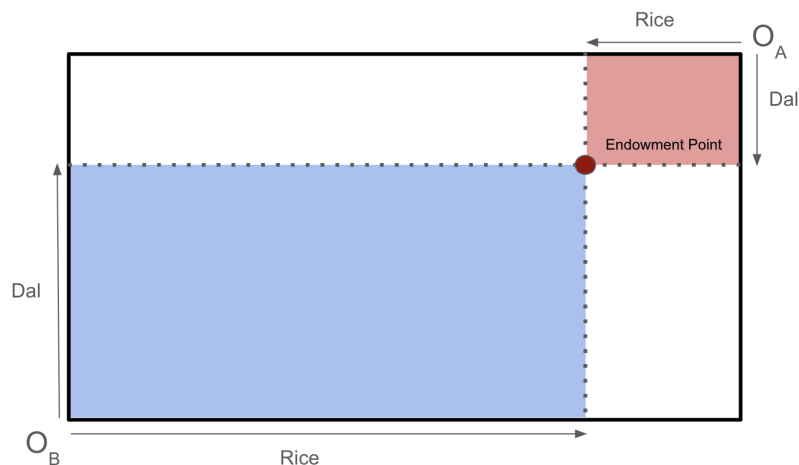


Figure 3.1: Is this an Edgeworth Box?

and hence have bargaining power in the economy.

It eventually comes down to "haves vs have-nots", or "efficiency vs equity". Efficiency would imply distribution of resources in a way that results in maximum potential for output, while equity would imply equal sharing of the output so generated by the economy.

Quantitative Easing? (this term was mentioned in the notes) **Quantitative easing (QE)** is when a central bank creates new money electronically and uses it to buy financial assets (like government bonds, corporate bonds, or mortgage-backed securities) from banks and other financial institutions. It's like minting money, but digitally.

Chapter 4

Lecture 2

Economics tries to solve the problem of scarcity, where scarcity addresses things like amenities, housing, sanitation, drinking water etc. The scarcity is in part aggravated as a consequence of scientific advancements. Developments in the field of science have resulted in an arrested mortality rate and a steady / increasing birth rate. This has created a situation where a larger population competes for nearly the same set of resources, implying increased scarcity.

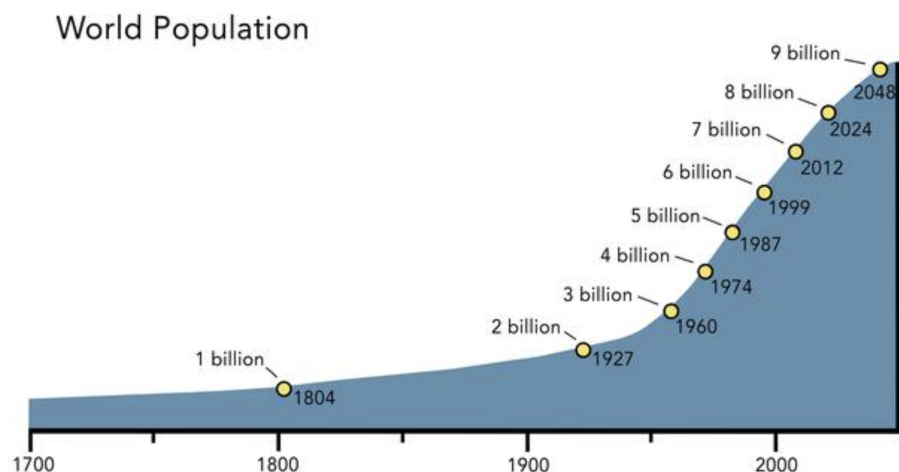


Figure 4.1: World Population over the years

APJ Abdul Kalam has a book called *India 2020: A Vision for the New Millennium*. His claim was that by 2020, India will have "**Demographic Dividend**". Demographic dividend is the economic growth potential resulting from a shift in a population's age structure, specifically when the working-age population is larger than the dependent population. As India would see a growth in young, capable and educated people joining the workforce, the workforce of

economies like China and Japan would start "greying", i.e. people who are in the workforce would age, and it would mostly be composed of older individuals. That is, while India's **dependency burden** would shrink, that of countries like Japan or China would increase in the future. Dependency burden can be thought of as a ratio of the number of people who depend on a provider for sustenance to the number of providers. (The high dependency burden promoted a shift from a joint family structure to a nuclear family).

Charlie Chaplin's *Modern Times* reflects the economics of the Industrial Revolution. Machines replaced manual labour, factors such as fatigue were eliminated from the workforce, and skill requirements changed from manufacturing goods by oneself to operating machines capable of manufacturing goods.

Amenities can't see the same explosive growth rate as population, even if the required monetary funds and technology are available. Solutions can't be implemented in short time. Consider the example of a bullet train. To set up a bullet train say between Mumbai and Surat, we have a compromise between laying tracks down the 'shortest path' between the locations, and the consequent relocation of people, markets, residential areas, deforestation etc. Even when the funds and technology are available, it has taken over 10 years for the project to be implemented (still to be open in 2027).

Indivisible Goods are public infrastructure or public goods. They are indivisible in the sense that their utility is 'boolean'. A half completed bridge is of no value. A train track with a missing rail in between has no utility either. An electrical transformer with unfinished wiring is useless. These add value / utility only if complete. On the other hand, some items have utility even in parts. For example, a pizza can be consumed across 3 days in parts, a shampoo can be used across months, i.e. it has value even without having to be used at once. This point on, *goods refers to public goods / infrastructure*, while *commodities refers to things such as pizza, shampoo etc.*

Price Floor and Price Ceiling from A Macroeconomic Perspective

- **Price Floor:** The price floor is a minimum price imposed by the government at which a commodity / good / service is sold. This price floor is kept above the equilibrium price of the market, so as to ensure a surplus in the market, thereby allowing for a 'buffer stock'. This buffer stock is kept to be redistributed via Public Distribution Systems to the poor, or to be distributed calamity affected regions. *Despite this surplus maintenance of say food and grains etc., we still have malnourishment?*
- **Price Ceiling:** The price ceiling is a maximum price imposed by the government at which a commodity / good / service is sold. This price ceiling is kept below the equilibrium price in the market, in order to ease off the load on people. An example would be *Rent Control Acts* imple-

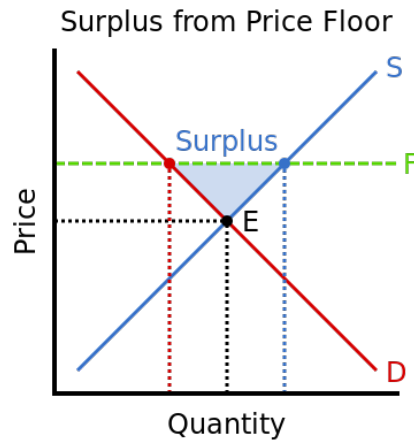


Figure 4.2: Price Floor

mented in various places, which implement a ceiling on the rental rates corresponding to the sizes of the property etc. This however, creates a shortage.

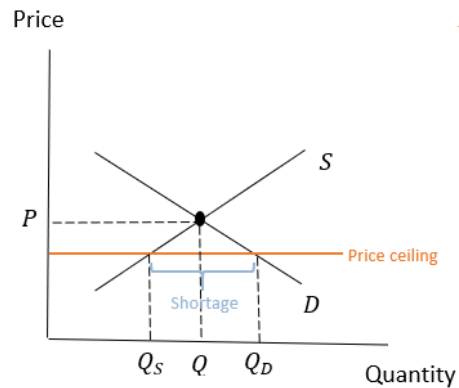


Figure 4.3: Price Ceiling

Price in a market tends to reflect quality. For example, if phone A is priced at \$ 100,000 and phone B is priced at \$ 40,000 then naturally, people tend to think phone A is of a higher quality than phone B. Likewise, a more expensive chocolate is considered to be of a higher quality than a lesser priced one. If a company offers multiple goods of differing prices (and hence qualities) to cater to its consumers, it is called **Vertical Differentiation** and the products are called vertically integrated products.

Inflation

Cases of Hyperinflation:

- **Bolivia (1983 - 1985)** : 23,000% increase in prices.
- **Germany (1923)** : 500% increase in prices per month, and real wages dropped to a third each month on average.
- **Zimbabwe**: It witnessed hyperinflation under the reign of self proclaimed president (practically a dictator) Robert Mugabe.
 - 1998**: 32%
 - 1999**: 59%
 - 2002**: 208%
 - 2006**: 1042.9%
 - 2008**: 100,000%

Zimbabwe had a demonination in notes of 100 trillion Dollars. Yugoslavia has had 5 million dinar and 500 billion dinar notes. Brazil saw hyperinflation (1967-94), China too (1948-55). India has fortunately never fallen prey to hyperinflation. This is because India has "Prudential Regulation" - the Reserve Bank of India (RBI) stays insulated from all kinda of criticism.

The main problem that inflation brings is soaring prices of goods and commodities, but with salaries that are unable to keep up. Venezuela witnessed a 1,300,000% inflation between 2000 and 2018. There were riots for toilet paper in supermarkets - even basic facilities weren't available. In order to evade high inflation rates, there needs to be a coherence in political direction, i.e. political stability. This can be visualised via the Production Possibilities Frontier. For example, if a government in one tenure bolsters only agriculture by increasing the price floor on agricultural produce, we would see the PPF shift upwards and outwards, but more so towards agriculture than towards industry, indirectly providing an incentive to move towards agriculture - since there aren't sufficient well paying opportunities in industries under such a circumstance. However, if in the next tenure, the government focuses on industries and tries to undo this imbalance by 'resetting' the price floors on agricultural produce and bolstering industries, there would now be more opportunities in the industrial sector, but a dearth of skill that can fill in those opportunities, since the public would have developed a skill for agriculture as a result of the previous incentive, prioritizing it over industrial skill.

Industrial Development is always on the basis of indivisible goods.

Some Terms and Indices:

- **Ease of Doing Business Index (EoDB)** : Self Explanatory.
- **Transparency International Index**: Like a measure for corruption.
- **Red Tapism**: Adherence to rigid or unnecessary 'formal' rules and procedures, creating delays in decision-making and operation while also increasing costs for businesses and people alike. It involves burdensome paperwork, redundant approvals, and slow processes that offer no added value and often lead to public dissatisfaction, reduced efficiency, and even corruption.

The Three Strains of Inflation

1. **Low Inflation**: Typically when the inflation rates are in single digits.
2. **Galloping Inflation**: When the inflation rate is in double digits, or early triple digits. Typically, a rate between 20% and 200% is considered galloping inflation.
3. **Hyperinflation**: Deadly rates of inflation, going up to millions or trillions of percentage.

In India, a rate of inflation over 6% is considered alarming by the RBI.

Mathematically, Inflation can be represented as:

$$\text{Inflation Rate} = \frac{\text{Price}_t - \text{Price}_{t-1}}{\text{Price}_{t-1}} \cdot 100\%$$

This "Price" however, is measured on a macroeconomic scale - it's an economy wide average price level, accounting for all goods, commodities and services. There are two price indices that we talk about - **Consumer Price Index** (CPI) and **Wholesale Price Index** (WPI). As a consumer, we are affected by CPI.

In very simple terms, **Inflation is the situation where too much money is chasing too few goods.**

Chapter 5

Lecture 3

5.1 Classification of Economies

We have seen hyperinflation in different case studies from around the world. There are some 'rich' and 'poor' countries across the globe — there is widespread economic inequality.

Economic systems can be classified on the basis of the following criteria:

- Who owns the Economic resources?
 - Capitalist
 - Socialist
 - Mixed
- What is the extent of Market Mechanisms?
 - Market Economy
 - Mixed Strategies
 - Planned Economy

In a **Capitalist** country, the major economic resources are owned by private entities and it is their decision on what to produce, how to produce and for whom to produce - not ideal for a democracy.

The opposite extreme of capitalism is **Socialism**, where all resources are 'state-owned'. However, this also tends to not work very efficiently - for e.g., in general, government provided resources doesn't give a very pleasant image to the mind, such as government hospitals.

Market mechanisms (natural demand and supply) may result in prices of goods and services which are out of range for a common man. Eg: Healthcare. If left to capitalism, we end up leaving the health of the public in the hands of private

entities. If left solely to the government, nobody tends to visit the healthcare centres due to ill-maintained hygiene, or unavailability of doctors etc.

The term socialist was added to the preamble of the Indian Constitution in 1976, the 42nd Amendment. The collapse of USSR allowed capitalists to point out that socialism had 'inherent flaws'. In 1989, the wall of Berlin was broken - West Germany (capitalist) and East Germany (socialist) ended up mixing together, but the capitalist sentiment prevailed. In the modern day, economies prefer not to abide by either extreme strategies. This leads to the classification of economies based on the extent of market mechanisms.

India is a **Market Economy**. We have a population of around 150 crores, while a population of less than 5 crores pays direct taxes. This is clearly not sufficient for the government to be able to fund and maintain to good standards the quality of all goods and services. So in Market Economies, the governments seek to operate via public-private cooperation. For example, the Government of India built multiple ports (Such as one in Kerala) alongside the Adani Group. Adani Ports manages almost all ports in India.

5.2 Public Investment

When we talk of public investments, we may have the following:

- Concentrated in the area of public infrastructure
- directed primarily towards agriculture
- directed towards Industrial development

Note that Public Investments are indivisible investments, implying that there is likely to be one particular field with massive investment, with the others witnessing significantly less funding. The natural question arises : **how to prioritize fields of investment?**

During India's first 5-year Plan post independence, after the 1st year there was widespread famine. Nehru invited Professor PC Mahalanobis to assist with the decision of distribution of funds. It was Professor Mahalanobis' suggestion that not all funding should be invested into agriculture - instead, industries needed to be funded too. This wasn't an easy decision to make - It was met by opposition from the people. Further, Industries can't be established everywhere simultaneously - it grows only in pockets. Even today, we have a few major industrial cities, while some areas remain backwards in the field of industry even today.

It's important to understand that say Bangalore is not the whole of Karnataka, or Kolkata is not the whole of West Bengal. While these cities have good industries, it's not sufficient to comment just on them when talking about



Figure 5.1: Professor Prasanta Chandra Mahalanobis

the people of the entire state.

At some point in the 1960s, South Korea and India were nearly equal in per capita income. But South Korea had a change in trade policies during the 1970s, as a consequence of which, it saw a massive boom in the per capita income. India however did not do so - there was the 1975 emergency too. As a consequence, India didn't have a boom in per capita income like South Korea.

For the production function, the **Bordered Hessian Determinant** must be positive.

$$\begin{vmatrix} 0 & f_L & f_K \\ f_L & f_{LL} & f_{LK} \\ f_K & f_{LK} & f_{KK} \end{vmatrix} > 0$$

Where 'f' is the production function, a function of labour (L) and capital (K). It is tied to the diminishing increase in return despite an increase in labour or capital.

Trade miracle

In the early 1950s, South Korea and India had comparable per capita incomes. However, in the 1960s, South Korea switched to export-led policies, resulting in economic transformation and GDP growth of 8.97% per year from 1960 to 2000.

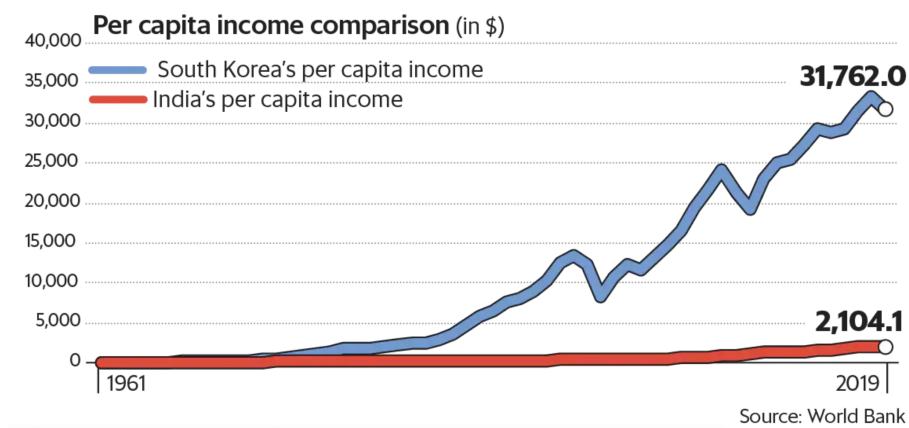


Figure 5.2: Per Capita income over the years, India vs South Korea

5.3 Market Economies Perform Better

- **Most goods and services are produced by private firms.**
- **The role of the government in the market is limited.** It may decide certain things like price floor on grains, price ceiling on rents etc. But the competition remains between the participating firms - the government doesn't intervene in the competition.
- **Quality and pricing of the products is auto-regulated,** because sub-par products, or products with unreasonably high prices will be outdone by higher quality or more rationally priced products from competitors. The products must be cutting edge.
- **Existence of Private Property Rights:**

Consider the following problem set-up. There's a hill, with a stream flowing down it. There is a washerman who lives atop the hill, and a fisherman who lives at the bottom. Both are unaware of each other's existence, however, both are known to the government. The washerman washes his clothes, due to which the stream water gets polluted upstream. Consequently, the fish die due to the pollution, because of which the fisherman now struggles, since he can't catch fish!

How should the government resolve this? The government can't say that

the washerman should stop using detergent to wash clothes - it's his livelihood. Nor can they ask the fisherman to change his occupation, the fisherman has his autonomy too.

The first time this problem was presented it was deemed 'trivial'. The solution suggested was to tax the washerman (the polluter), and subsidise the fisherman (the one affected by the pollution). This was, however, very quickly and vehemently dismissed. **Is paying a tax equivalent to buying the right to pollute?** Further, to fulfil the **tax**, the washerman will **wash more clothes**, causing more pollution of the water. It is the same for the fisherman too - if you were to ask him to sit at home, while the government provides for him via subsidies, it would be opposed too. The **fisherman** is an able-bodied skilled civilian, who **wants to work**.

The solution to this problem (it won **Ronald Coase** the Nobel Prize) comes from the existence of private property rights. People tend to maintain their 'own' property. For example, people don't litter in their own house. They are more likely to throw trash on the roads or in the public than on the floor of their own house (claim based on observation). People are protective of private property. If the same is applied to the washerman, where the upstream part of the stream is 'leased' to the washerman for a span of time, the washerman would naturally try to maintain it. Same with the fisherman.

This same thing also applied in Operation Sindoor. USA was trying to resolve the conflict between India and Pakistan - and was instantly questioned by the conflicting parties. **Bilateral problems can't be resolved by a third intervening party.**

The Coase Theorem: If property rights are well-defined and transaction costs are zero (or negligible), parties will bargain privately to reach an efficient allocation of resources, regardless of the initial allocation of rights.

- **Right of Freedom of Enterprise and Choice:** The government can't bar the production of a type of goods / a variant of a type of goods - this would lead to red-tapism and 'License-Permit-Raj'.

5.4 Characteristics of Developing Nations

The following are common characteristics of 'developing' nations:

- Low Standards of Living
- Low Productivity
- High rates of population growth and dependency burden

- Substantial dependence on agricultural production and primary product exports
- Prevalence of **imperfect** markets

Dependency burden, as mentioned earlier, is the ratio of dependent young and old to population of working age, as a country goes through demographic transition.

Imperfect Markets

Very simply put, an imperfect market is a market which deviates from the conditions of perfectness. In a perfect market, we have

$$\text{Marginal Cost (MC)} = \text{Price (P)}$$

. Consider the following derivation: In a perfect market, we have $MC = P$. In any general market, the following holds:

$$\text{Total Revenue (TR)} = \text{Marginal Revenue (MR)} = \text{Price(P)} \times \text{Quantity(Q)}$$

$$\Rightarrow \frac{d(TR)}{d(Q)} = P + Q \frac{dP}{dQ}$$

In a perfect competition, no firm has 'market power', i.e. the dP/dQ term is zero. Which means, $MR = P = MC$. However, in imperfect competition, we may write this expression as:

$$\begin{aligned} MR &= P + Q \frac{dP}{dQ} \\ \Rightarrow MR &= P \cdot \left(1 + \frac{QdP}{PdQ} \right) \\ \Rightarrow MR &= P \cdot \left(1 - \frac{1}{-\frac{dQ/Q}{dP/P}} \right) \\ \Rightarrow MR &= P \cdot \left(1 - \frac{1}{-\frac{dQ/Q}{dP/P}} \right) \\ \Rightarrow MR &= P \cdot \left(1 - \frac{1}{|E_d|} \right) \end{aligned}$$

where E_d is the Price Elasticity of Demand. If we now try to set $MR = MC$ (condition of maximum profit), we get $P > MC$. There is some tendency of Monopolistic / Oligopolistic competition in developing nations, where firms face

a downward sloping demand curve. When P exceeds MC , we have a 'markup', which reflects monopolistic power in the economy - since this markup or premium is set by the Entrepreneurs selling the goods and services.

Mexico, Asia, South America, Africa come under developing parts of the world, while North America, Europe, Russia, Japan, Korea, Australia etc come under the developed parts of the world.

5.5 The Vicious Cycle of Poverty

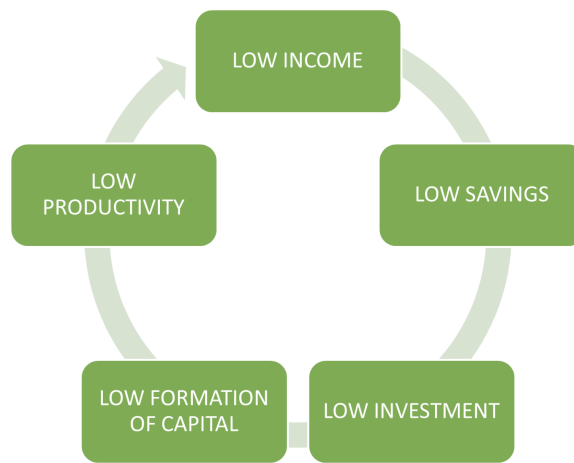


Figure 5.3: The Vicious Cycle of Poverty

The figure is self explanatory.

The statement "**A country is poor because it is poor**" by **Ragnar Nurkse** is a statement that aligns with the concept of the vicious cycle of poverty. This idea explains how low income and low productivity levels in developing countries reinforce each other, trapping the country in poverty.

A 'poor' country lacks the resources such as capital, infrastructure, and education that are needed to increase productivity. Low productivity means low income. Low income means low savings and investment. Low investment means the country cannot build the resources necessary for higher productivity, thus perpetuating the cycle of poverty.

5.6 Prices: Flexible and Sticky

”Sticky” (in this context) means something that adjusts sluggishly to the change in demand or supply. In general, we have an equation:

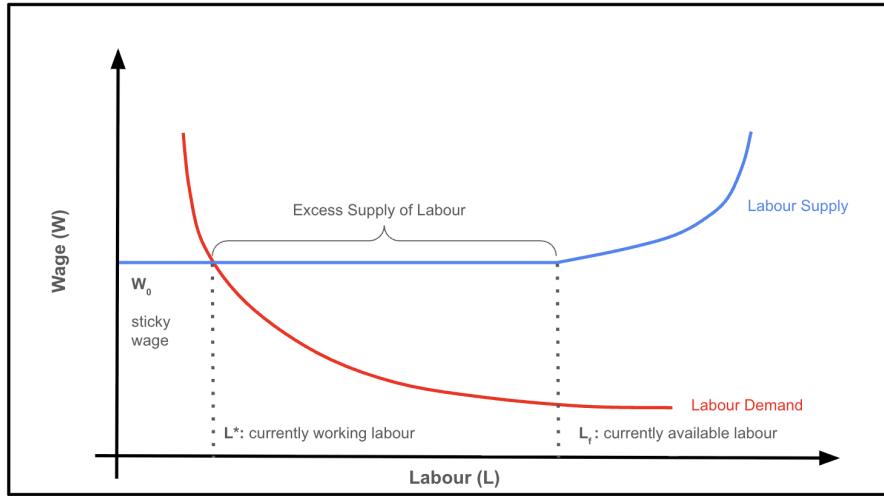


Figure 5.4: Sticky Wage - Demand and Supply

$$\frac{d(\text{Price})}{d(\text{time})} = \lambda \times \{D(P) - S(P)\} \quad \lambda > 0$$

Where $D(P)$ and $S(P)$ are the demand and supply at price 'P' respectively.

- If $D(P) > S(P)$, we have a shortage, hence price should naturally increase.
- If $D(P) < S(P)$, we have an excess, hence price should drop.

λ is called the rate of adjustment. But, this fails if the price is sticky! Prices are generally sticky in the short run (As a result of contracts - say daily wage of a mason is Rs. 625 a day, and this contract holds for 1 year — this has made the wage sticky).

Some consequences of Stickiness are:

- Sometimes demand may exceed supply, resulting in unemployment (The sticky wage situation discussed above).
- Sometimes firms can't sell all goods they produce (sticky price situation).

5.7 Circular Flow of Economy

This is the circular flow diagram discussed in micro-economics too.

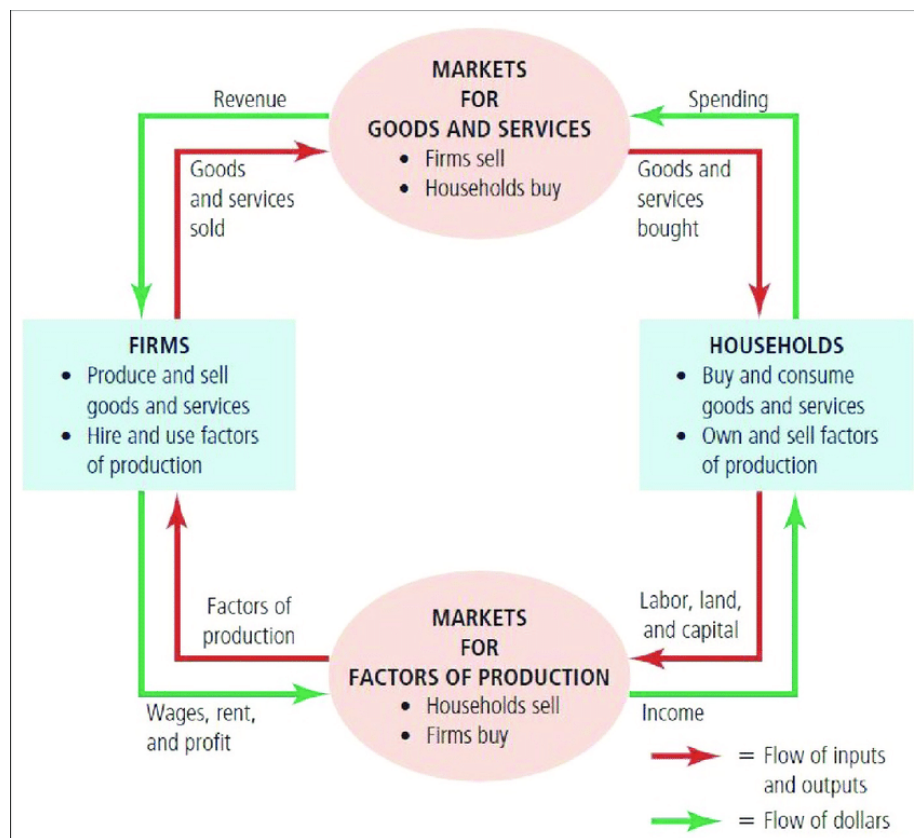


Figure 5.5: Circular Flow Diagram

Chapter 6

Lecture 4

6.1 The Two Sector Model

This is one of the simplest models in economics. The two sectors involved are **households** and **firms or businesses**. The way this model works is: Households provide factors of production that are sought by the firms:

- Labor
- Land
- Capital
- Entrepreneurship

In return, firms pay the households through income via salaries, dividends from profits, interest payments etc. If we were to also include the government and the foreign sector, then we have a four-sector model, where firms and household pay taxes to the government, and the government provides subsidies, exports and imports. The foreign sector contributes in research, development and defense.

6.2 National Income Accounting

6.2.1 The Value Added Approach

$$Profit = Total\ Value - Total\ Cost$$

Where the total cost is the total cost of the factors of production. The value added approach is also called the product approach.

The Value Added of a producer is the value of its output minus the value of its intermediate inputs it purchases from other producers.

Output is demand determined.

What is the aggregate demand in our economy? Aggregate is calculated based

The Concept of Value Added

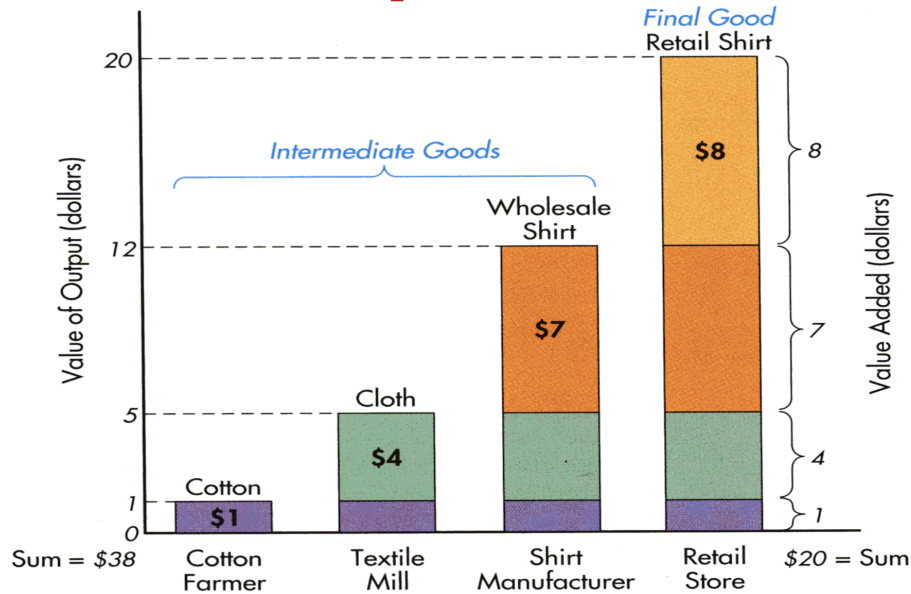


Figure 6.1: The Value Added Approach (from slides)

off all the stakeholders in the economy.

Gross Domestic Product (GDP): Goods produced within the geographical boundaries of India under the current market price (Second-hand products are not counted). Per capita GDP would be

$$\text{Per Capita GDP} = \frac{\text{GDP}}{\text{Total Population}}$$

Gross National Product (GNP): $\text{GNP} = \text{GDP} + \text{NFIA}$.

NFIA stands for Net Factor Income from Abroad. It can be positive, negative or 0. For India, it is negative.

Non-Market Productive Activities - Important activities that have no well-defined market. Examples could include a variety of household work (such as cooking, cleaning etc).

6.2.2 Green Income

Green Income refers to income that is earned from environmentally sustainable activities or adjusted income that accounts for environmental impacts.

It is viewed in two ways:

- It includes the income people or businesses earn by doing work that protects the environment or uses natural resources sustainably.
 - Earnings from solar or wind energy production
 - Income from organic farming or sustainable agriculture
 - Jobs in recycling industries or waste management
 - Eco-tourism guides or forest conservation workers
 - Manufacturing of electric vehicles, solar panels, biodegradable products
- **Environmentally Adjusted National Income (Green National Income / Green GDP)** This is an economic concept used by governments and economists. It means calculating a nation's income after subtracting the cost of environmental damage and natural resource depletion. So instead of only counting factories, production, and profits, it also considers:
 - Pollution of air, land, and water
 - Deforestation and loss of biodiversity
 - Depletion of minerals, coal, oil, forests
 - Cost of cleaning and restoring the environment

Chapter 7

Lecture 5

7.1 National Income accounting (continued)

Supply of Output / GDP (Y):

$$Y = C + I + G + (X - M)$$

The term $C+I+G+(X-M)$ is called the Aggregate Demand of the Economy. The individual terms are:

- **C**: Consumption (Household Sector)
- **I**: Private Gross Investment (Businesses or Firms (Business Sector))
- **G**: Government Expenditure (Government Sector)
- **X**: Exports (Foreign Demand for Indian Products)
- **M**: Imports (Home Demand for Foreign Products)

Note that firm investments are considered in the calculation, and not stock. Also, the concept of investments in stock by households is not considered in economics.

7.1.1 Depreciation

It is defined as the **wear and tear of a capital stock**.

$$Net\ Investment = Gross\ Investment - Depreciation$$

Depreciation can be positive or negative.

Net Domestic Product (NDP) is GDP - Depreciation. Likewise, Net National Product (NNP) is GNP - Depreciation. Analysing on a household level:

$$Y = C + I + G + (X - M)$$

$$Y_0 = C + I + (G + T_R - T_X) + (X - M)$$

Y_0 is called the Disposable Income. T_R is called the Transfer Payment. Eg: Old Age Pensions, Social Security Payments, Dole (unemployment benefits). T_X is taxes. Disposable income can be spent or saved (not invested! Investments considered in economics are by businesses). Also,

$$Y_0 = C + S$$

where C is consumption and S is savings. Equating the 2 expressions for Y_0 , we get:

$$S - I = (G + T_R - T_X) + (X - M)$$

- S - I is called the savings-investment gap.
- $G + T_R$ is the government's expenditure.
- T_X is the government's income.
- if $G + T_R - T_X$ is > 0 , we say there's a government budget deficit, else if < 0 , it's a government budget surplus.
- $X - M > 0$ is a trade surplus, < 0 is a trade deficit.

A closed economy is one which has no economic interactions with the rest of the world - i.e. no imports or exports. India became an open economy in June 1991.

In a closed economy without a government, we have $Y = C + I$ and $Y = C + S$, which gives us the income and savings equality. When we have an open economy, we revert to the initial equation

$$S - I = G + T_R - T_X + (X - M)$$

S is a leakage, and I is an injection. $G + T_R - T_X$ is the domestic component, and $X - M$ is the foreign component. S no longer equals I. Since the money represented by S is out of the system, it can't be loaned to firms to make investments. C is the most stable component of aggregate demand anywhere in the world. That's because People must spend on essentials like food, clothing, housing, medicines, electricity, etc. Even if income falls or the economy slows, people still consume these goods. They may reduce luxury spending a bit, but basic consumption remains steady. Also, people prefer a smooth lifestyle — this is called the “Consumption Smoothing” behavior. If income falls → people use savings or borrow to maintain consumption. If income rises → they save the extra instead of spending it all. So, consumption doesn't fluctuate as wildly as income.

(In general, real means inflation adjusted)

According to **Keynes' Consumption Function** in its linear form(Given by Sir J.M. Keynes - considered the 'God' or 'Founding Father' of modern macroeconomics):

$$C = a + bY$$

where:

a = autonomous consumption (basic spending even at zero income).

b = MPC (marginal propensity to consume), usually stable in the short term.

bY is called the Induced Consumption term (Spend and Earn) This formula shows consumption depends on income but not dramatically, because of the stable values of a and b.

Chapter 8

Lecture 6

GDP can be evaluated at the current market prices. When we say nominal GDP, it is not inflation adjusted, and directly uses the current market prices. When we talk about the Real GDP though, it is inflation adjusted, and based off the prices of a 'base year'. The choice of base year is done carefully - it shouldn't be a year which witnessed some large-scale disaster like COVID-19, or a war etc - it should be a normal year. The base year is updated regularly. In short:

- Nominal GDP: GDP at current prices
- Real GDP: GDP at constant prices
- For the 'base year', Nominal GDP = Real GDP

The Index or Rate of inflation is calculated as

$$\frac{Normal \text{ GDP}}{Real \text{ GDP}} \cdot 100\%$$

Price indices such as consumer price index and wholesale price index (CPI and WPI) have been mentioned earlier. CPI is relevant to us, and can be viewed as 3 components: CPI_R , CPI_U , and CPI_{IW} . (R - Rural, U - Urban, IW - Industrial Work).

In the short run, we know that unemployment varies inversely as the inflation rate.

$$Growth \text{ of Output} = \frac{GDP_t - GDP_{t-1}}{GDP_{t-1}} \times 100\%$$

Low Involuntary Unemployment - people wish to be employed, but are unable to get employment. In an ideal situation, inflation rates should be low and stable. RBI has tried to maintain this rate at 4-6%.

There's a **Policy Trade off**. If there's a high rate of employment (desirable),

it results in a high income level, consequently a high demand, increasing prices, which means an increase in inflation (which is undesirable!) **Output responds to demand, but with a lag.**

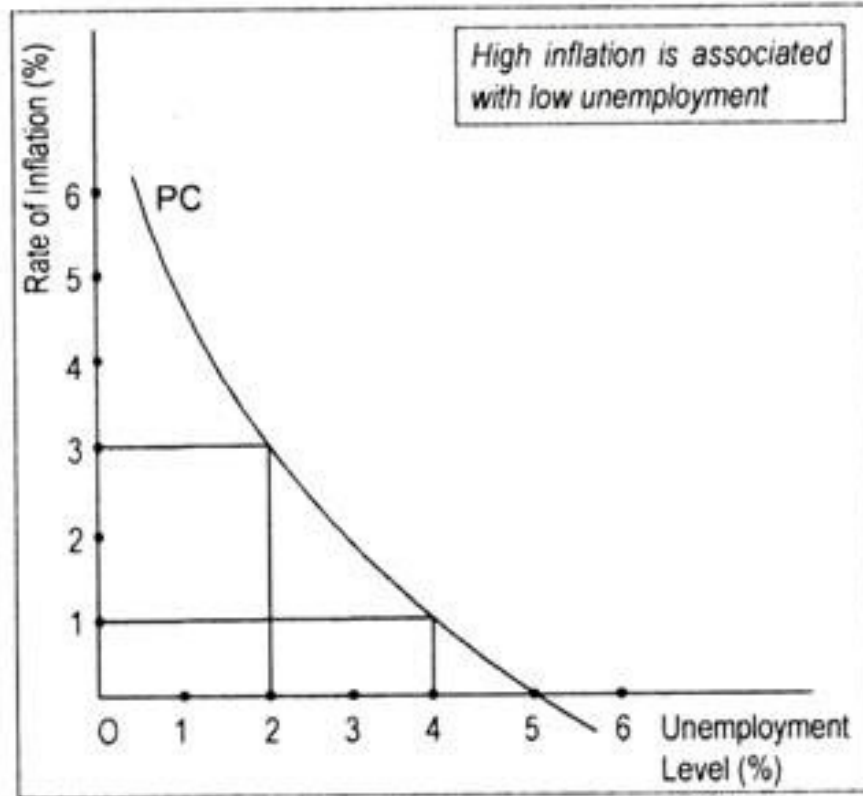


Fig. 4.5: Phillips Curve

Figure 8.1: The Philips Curve - given by Arthur Philips

The natural rate of unemployment is where the philips curve intersects with the unemployment level axis.

$$\text{Current Real Consumption } (C) = f(Y_d) \quad f' > 0$$

where Y_d is the current **real** disposable income. In the linear expression $c = a + bY$, we have $a > 0$ and $0 < b < 1$. Like MPC, we define APC (Average Propensity to Consume) as the consumption per unit of disposable income.

$$APC = \frac{c}{Y_d} = \frac{a}{Y_d} + b$$

$APC - MPC > 0$. There is no role of interest rate in determining household consumption. As Y_d increases, APC decreases. Simon Kuznets once wrote a paper on why APC doesn't decrease with an increase in income (1942), based on data from 1869 - 1929. It was debated until 1944. It was called **Kuznet's Puzzle**.

Chapter 9

Lecture 7

If taxes decrease, then Y_d and consequently disposable income and consumption both increase. Consumption is the largest component of both aggregate demand and the GDP. We have the tax function in macroeconomics:

$$T = T_0 + eY \quad 0 < e < 1$$

if T_X increases, then consumption and aggregate demand, and as a result output decreases (output responds to demand).

In a capitalist closed economy without the role of a government, we have $Y_d = Y$. APC varies inversely as Y , while MPC remains constant w.r.t Y (at b). We have the observation that $MPC_{rich} < MPC_{poor}$, since rich people already have a lot of facilities. Consequently, $MPS_{rich} > MPS_{poor}$ (MPS is the Marginal Propensity to Save) i.e. rich people save more.

$$C = a + bY_d \quad - (i)$$

$$Y_d = C + S \quad - (ii)$$

$$\Rightarrow S = (1 - b)Y_d - a$$

This equation is called the Keynesian saving function. MPS is the partial derivative of S w.r.t $Y_d = 1 - b$, which is more than 0 (since b is between 0 and 1). Also,

$$Y_d = C + S$$

$$\Rightarrow 1 = MPS + MPC$$

(if we take a derivative w.r.t Y_d) And

$$1 = APC + APS$$

(if we divide both sides by Y_d).

Prior to 1929, it was believed that interest rates contribute to consumption. However, this was later disproved by Keynes's **Absolute Income Hypothesis**. The Absolute Income Hypothesis states that consumption (C) depends primarily on the absolute or current level of income (Y), not on past or future income.

The **Secular Stagnation Hypothesis** is an economic theory that suggests that an economy can experience prolonged periods of very low growth, low demand, low interest rates, and high unemployment—even when there is no recession or crisis. Secular stagnation means long-term (secular = long-lasting) economic stagnation where the economy's natural ability to grow and generate full employment weakens.

A. Modigliani and R. Brumberg argued that Keynes' theory was wrong. It was later shown by the MIT that Keynes is right on a short timescale. In the long run, $C \neq f(\text{income})$, but $C = f(\text{income}, \text{wealth})$.

A desirable consumption pattern is the smoothest possible one. (A consumption pattern refers to the way a person's consumption changes over their lifetime or over periods of time. It's basically a plan or profile of spending: how much you consume in youth, middle age, and old age.) A smooth pattern is better because people prefer stability over their lifetime.

We get the equation:

$$C = \frac{W + RY}{T}$$

Where:

- W : Wealth accumulated - real estate, stocks, investment
- R : Working years
- Y : Income
- T : Life Expectancy.

$$\frac{C}{Y} = \frac{\alpha W}{Y} + \beta$$

β is the MPC term, which was accounted for by Keynes. But it didn't account for the first term.

Income can be seen as composed of 2 major components - a stable income called the permanent income (job), and transfer income (jackpot). Another person made a claim that in the consumption ideologies discussed earlier, nobody has discussed transfer income.

Chapter 10

Lecture 8

Let us consider the following models:

- **Keynes' AIH (Absolute Income Hypothesis):** Consumption depends only on the current (absolute) level of disposable income (no interest contribution).
 $C = f(Y_d)$, $f' > 0$
- **Modigliani's LCH (Life Cycle Hypothesis):** Consumption depends on expected lifetime income, not just current income. People try to smooth consumption over their life.
 $C = h(Y_d, W)$, $h_1 > 0$, $h_2 > 0$
(h_1 and h_2 are derivatives w.r.t Y_d and W .)
- **Milton Friedman's PIH (Permanent Income Hypothesis):**
Consumption depends on a person's permanent (long-term expected) income, not just their current income.
 $C = \phi(Y_p)$, $\phi' \geq 0$

if $Y > AD$ (aggregate demand), we have a situation of unintended inventory accumulation. **Excise Tax / Duty** is a type of indirect tax levied on the production, sale, or consumption of specific goods or services. Unlike income tax, it's collected per unit or as a percentage of price, and it's usually included in the price paid by the consumer.

10.0.1 The Income Method

GDP_{FC} = GDP calculated based off factor costs.

GDP_{MP} = GDP calculated based off market prices.

$$GDP_{MP} = GDP_{FC} + \text{net Indirect Tax}$$

where net indirect tax = Indirect tax - subsidies. Now, as we know, $Y = C + I$ and C is a function of Y . So if I goes up by 1 unit, Y goes up by one unit, which later reflects in C going up by C' units (where C' is the derivative of C w.r.t. Y , a.k.a. the MPC.), which would mean Y went up by another C' units, hence C would go up by $(C')^2$ units, and so on. Therefore, we have:

$$\Delta Y = (1 + C' + (C')^2 + \dots)\Delta I = \frac{\Delta I}{1 - C'}$$

$$\Rightarrow \frac{\Delta Y}{\Delta I} = \frac{1}{1 - MPC}$$

This term $\Delta Y/\Delta I$ is called the Keynesian Autonomous Investment Multiplier. It is always greater than 1 (since MPC is always between 0 and 1). Also, since $MPS + MPC = 1$, we have:

$$\frac{\Delta Y}{\Delta I} = \frac{1}{MPS}$$

This is why savings is considered a 'leakage' (see pg 25).

10.0.2 Deficit Spending

Franklin D. Roosevelt implemented a 'new deal'. Since we have:

$$Y_d = C + I + (G + T_R - T_X) + (X - M)$$

An increase in G will increase the budget deficit. And as demonstrated earlier with I , if G is increased by 1 unit, Y increases by 1 unit, which becomes a positive feedback loop between C and Y , and we get the result:

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1 - MPC} \quad 0 < MPC < 1$$

Here, ΔG is called the Government Expenditure Multiplier, or Keynesian Fiscal Multiplier.

The term output gap is the difference between the actual output and the potential output. It can be positive or negative.

$$\text{Output Gap} = \text{Actual Output} - \text{Potential Output}$$

An observed example is : if actual output growth exceeds potential output by 2.25% then unemployment drops by 1%.